Project: Worked examples
Title: Bolted joint (simplified VDI 2230)
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1. Scope and method

One bolt clamping members under an axial working load: the preload from the tightening torque with its scatter (or as measured), the stiffness of bolt and members, how much of the working load the bolt sees, the residual clamp after embedding, separation, yield under preload plus working load, the thread engagement needed against stripping, and the fatigue amplitude.
$A_s = (\pi/4)\cdot((d_2 + d_3)/2)^2$, $d_2 = d - 0.6495\,P$, $d_3 = d - 1.2269\,P$. Nut factor: $F_M = M_A/(K\,d)$. Friction: $F_M = M_A/(0.16\,P + 0.58\,d_2\,\mu_G + 0.5\,D_{Km}\,\mu_K)$. $F_{M,min} = F_M/\alpha_A$. $\delta_S = (0.4\,d + 0.4\,d)/(E_S\,A_N) + l_k/(E_S\,A_s) + 0.5\,d/(E_S\,A_{d3})$. Members: two frusta, $k = 0.5774\,\pi\,E\,d\,/\,\ln[((1.155\,t + d_w - d_h)(d_w + d_h))/((1.155\,t + d_w + d_h)(d_w - d_h))]$. $\Phi = n\,k_S/(k_S + k_P)$. $F_Z = f_Z\,(k_S\,k_P)/(k_S + k_P)$. Given preload: $F_M = F_V$; k_S, k_P and A_s may be given (measured, FE, a fastener record) instead of derived. $F_{KR} = \max(F_{M,min} - (1 - \Phi)\,F_A - F_Z, 0)$. $F_{S,max} = \max(F_M + \Phi\,F_A, F_A)$: once the joint opens the bolt carries the whole working load. $\sigma_{red} = \sqrt{(\sigma_z^2 + 3\,(0.5\,\tau)^2)}$. $\sigma_a = (F_{S,max} - F_M)/(2\,A_s)$. $m_{min} = S\,F_{S,max}/(0.5\,\pi\,d\,\tau_B)$.
Ref: VDI 2230 Part 1:2015 §5.1–§5.5 — Systematic calculation of highly stressed bolted joints — calculation steps R0–R13
Ref: ISO 898-1:2013 §9.1.6 — Tensile stress area of the thread
Ref: Shigley's Mechanical Engineering Design §8-5, §8-7 — Joint stiffness (frusta); preload and torque

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
d	Nominal thread diameter	12	mm	drawing: DRG-FLG-02 rev A	3 … 64 mm
P	Thread pitch	1.75	mm	spec: ISO 261 coarse pitch	0.35 … 6 mm
R_p02	Bolt 0.2 % proof (yield) strength	640	MPa	spec: ISO 898-1 class 8.8	200 … 1200 MPa
E_S	Bolt Young's modulus	205	GPa	spec: steel	60 … 250 GPa
sigma_A	Bolt endurance limit (stress amplitude)	50	MPa	assumption: Rolled-before-heat-treatment M12, VDI 2230 order of magnitude.	10 … 200 MPa
l_k	Clamp length	40	mm	drawing: DRG-FLG-02 rev A	1 … 2000 mm
d_h	Hole diameter	13	mm	drawing: DRG-FLG-02 rev A	1 … 100 mm
d_w	Head / washer bearing diameter	18	mm	spec: ISO 4762 head bearing	2 … 200 mm
E_P	Clamped members' Young's modulus	205	GPa	spec: steel	5 … 250 GPa
n	Load introduction factor (0 = at the clamped faces, 1 = under the head)	0.5		assumption: Load introduced at mid-thickness.	0 … 1

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
f_Z	Embedding (settling) loss	11	µm	assumption: Three interfaces, machined.	0 … 100 µm
method	Preload method	friction		spec: Assembly procedure AP-7	any
M_A	Tightening torque	80	N*m	spec: Assembly procedure AP-7	0.1 … 10000 N*m
K	Nut factor (nut-factor method)	0.2		assumption: Not used by the friction method.	0.1 … 0.35
mu_G	Thread friction coefficient	0.12		assumption: Lightly oiled zinc-plated thread.	0.06 … 0.25
mu_K	Head / nut bearing friction coefficient	0.12		assumption: Lightly oiled head bearing.	0.06 … 0.25
alpha_A	Tightening scatter factor (F_M,max / F_M,min)	1.6		assumption: Torque wrench, VDI 2230 Table A8 order.	1 … 4
F_A	Axial working load (per bolt)	10	kN	spec: LOAD-CASE-3	0 … 5000 kN
F_KR_req	Required residual clamp force	5	kN	spec: SPEC-FLG §6 sealing	0 … 5000 kN
tau_B	Nut / internal-thread material shear strength	400	MPa	assumption: Nut material shear strength.	50 … 1200 MPa
m_eff	Thread engagement length provided	12	mm	drawing: DRG-FLG-02 rev A	1 … 200 mm
S_strip	Safety factor on stripping	1.25		assumption: Stripping safety factor.	1 … 3

3. Assumptions

A1. The joint is concentrically clamped and concentrically loaded; the members behave as two 30° frusta in series. (Eccentric clamping and loading (VDI 2230 full procedure) are outside this sheet.)
A2. Thread torsion is taken at 50 % in the equivalent stress for the working state. (Common VDI 2230 practice (k_τ = 0.5).)
A3. The working load is a pulsating 0…F_A load; the fatigue amplitude is Φ F_A / (2 A_s). (Compare with the bolt's endurance limit as entered.)
A4. Thread stripping uses a shear area of half the pitch per unit length on the nominal diameter. (A conservative simplification of ISO 898-2 / VDI 2230 §5.5.5.)
A5. sigma_A (Bolt endurance limit (stress amplitude)) is taken as 50 MPa. (Rolled-before-heat-treatment M12, VDI 2230 order of magnitude.)
A6. n (Load introduction factor (0 = at the clamped faces, 1 = under the head)) is taken as 0.5. (Load introduced at mid-thickness.)
A7. f_Z (Embedding (settling) loss) is taken as 11 µm. (Three interfaces, machined.)
A8. K (Nut factor (nut-factor method)) is taken as 0.2. (Not used by the friction method.)
A9. mu_G (Thread friction coefficient) is taken as 0.12. (Lightly oiled zinc-plated thread.)
A10. mu_K (Head / nut bearing friction coefficient) is taken as 0.12. (Lightly oiled head bearing.)

A11. alpha_A (Tightening scatter factor (F_M,max / F_M,min)) is taken as 1.6. (Torque wrench, VDI 2230 Table A8 order.)
A12. tau_B (Nut / internal-thread material shear strength) is taken as 400 MPa. (Nut material shear strength.)
A13. S_strip (Safety factor on stripping) is taken as 1.25. (Stripping safety factor.)

4. Working

[23] d_2 = d - 0.6495 * P — ISO 898-1 §9.1.6
= 12 mm − 0.6495 × 1.75 mm
= 10.863 mm
Pitch diameter

[24] d_3 = d - 1.2269 * P — ISO 898-1 §9.1.6
= 12 mm − 1.2269 × 1.75 mm
= 9.8529 mm
Minor diameter

[25] A_s = pi / 4 * ((d_2 + d_3) / 2)^2 — ISO 898-1 §9.1.6
= π / 4 × ((10.863 mm + 9.8529 mm) / 2)^2
= 84.266 mm²
Tensile stress area

[26] A_N = pi / 4 * d^2
= π / 4 × (12 mm)^2
= 113.1 mm²
Nominal shank area

[27] A_d3 = pi / 4 * d_3^2
= π / 4 × (9.8529 mm)^2
= 76.247 mm²
Minor-diameter area

[28] D_Km = (d_w + d_h) / 2
= (18 mm + 13 mm) / 2
= 15.5 mm
Mean bearing diameter under the head

[29] F_M = M_A / (0.16 * P + 0.58 * d_2 * mu_G + 0.5 * D_Km * mu_K) — VDI 2230 §5.4.3
= 80 N·m / (0.16 × 1.75 mm + 0.58 × 10.863 mm × 0.12 + 0.5 × 15.5 mm × 0.12)
= 40.69 kN
Assembly preload (friction)

[30] F_M_min = F_M / alpha_A — VDI 2230 R2
= 40.69 kN / 1.6
= 25.431 kN
Minimum preload after scatter

[31] sigma_M = F_M / A_s
= 40.69 kN / 84.266 mm²
= 482.87 MPa
Preload stress

[32] M_G = F_M * (0.16 * P + 0.58 * d_2 * mu_G) — VDI 2230 §5.4.3
= 40.69 kN × (0.16 × 1.75 mm + 0.58 × 10.863 mm × 0.12)
= 42.158 N·m
Thread torque under preload

[33] W_p = pi / 16 * d_3^3
= π / 16 × (9.8529 mm)^3
= 187.81 mm³
Polar section modulus at the minor diameter

[34] tau_M = M_G / W_p
= 42.158 N·m / 187.81 mm³
= 224.47 MPa

Thread torsional stress under preload

[35] sigma_red_M = sqrt(sigma_M^2 + 3 * tau_M^2) VDI 2230 R8

= sqrt((482.87 MPa)^2 + 3 × (224.47 MPa)^2)

= 619.94 MPa

Equivalent stress at assembly (full torsion)

[36] nu_M = sigma_red_M / R_p02

= 619.94 MPa / 640 MPa

= 0.96866

Utilisation of proof strength at assembly

[37] delta_S = 0.4 * d / (E_S * A_N) + l_k / (E_S * A_s) + 0.5 * d / (E_S * A_d3) + 0.4 * d / VDI 2230 Sh 1

= 0.4 × 12 mm / (205 GPa × 113.1 mm²) + 40 mm / (205 GPa × 84.266 mm²) + 0.5 × 12 mm / (205 G

76.247 mm²) + 0.4 × 12 mm / (205 GPa × 113.1 mm²)

= 0.0000031135 m/kN

Bolt resilience (head, shank, engaged thread, nut)

[38] k_S = 1 / delta_S

= 1 / 0.0000031135 m/kN

= 321190 kN/m

Bolt stiffness

[39] t_f = l_k / 2

= 40 mm / 2

= 20 mm

Frustum height (two equal frusta)

[40] k_f = 0.5774 * pi * E_P * d / ln(((1.155 * t_f + d_w - d_h) * (d_w + d_h)) / ((1.155 * t

(d_w - d_h)))

Shigley eq. 8-20

= 0.5774 × π × 205 GPa × 12 mm / ln((1.155 × 20 mm + 18 mm − 13 mm) × (18 mm + 13 mm) / ((1.1

mm + 18 mm + 13 mm) × (18 mm − 13 mm)))

= 3815600 kN/m

Stiffness of one frustum

[41] k_P = k_f / 2 Shigley eq. 8-21

= 3815600 kN/m / 2

= 1907800 kN/m

Clamped members' stiffness (two frusta in series)

[42] Phi = n * k_S / (k_S + k_P) VDI 2230 R3

= 0.5 × 321190 kN/m / (321190 kN/m + 1907800 kN/m)

= 0.072047

Load factor

[43] F_Z = f_Z * (k_S * k_P) / (k_S + k_P) VDI 2230 R4

= 11 µm × 321190 kN/m × 1907800 kN/m / (321190 kN/m + 1907800 kN/m)

= 3.024 kN

Preload loss from embedding

[44] F_SA = Phi * F_A

= 0.072047 × 10 kN

= 0.72047 kN

Additional bolt load from the working load

[45] F_PA = (1 - Phi) * F_A

= (1 − 0.072047) × 10 kN

= 9.2795 kN

Load relieved from the members

[46] F_KR = max(F_M_min - F_PA - F_Z, 0 kN) VDI 2230 R7

= max(25.431 kN − 9.2795 kN − 3.024 kN, 0 kN)

= 13.128 kN

Residual clamp force (minimum; zero once the joint opens)

[47] F_S_max = max(F_M + F_SA, F_A)
= max(40.69 kN + 0.72047 kN, 10 kN)
= 41.41 kN
Maximum bolt load (maximum preload + additional load; the whole working load once the joint opens)
[48] sigma_z_max = F_S_max / A_s
= 41.41 kN / 84.266 mm²
= 491.42 MPa
Maximum bolt tensile stress
[49] sigma_red_B = sqrt(sigma_z_max^2 + 3 * (0.5 * tau_M)^2) VDI 2230 R8
= sqrt((491.42 MPa)^2 + 3 × (0.5 × 224.47 MPa)^2)
= 528.47 MPa
Equivalent stress in the working state (50 % torsion)
[50] u_yield = sigma_red_B / R_p02
= 528.47 MPa / 640 MPa
= 0.82574
Yield utilisation in the working state
[51] sigma_a = (F_S_max - F_M) / (2 * A_s) VDI 2230 R9
= (41.41 kN − 40.69 kN) / (2 × 84.266 mm²)
= 4.2749 MPa
Fatigue stress amplitude (pulsating 0…F_A load)
[52] u_fatigue = sigma_a / sigma_A
= 4.2749 MPa / 50 MPa
= 0.085499
Fatigue utilisation
[53] m_min = S_strip * F_S_max / (0.5 * pi * d * tau_B) VDI 2230 §5.5.5
= 1.25 × 41.41 kN / (0.5 × π × 12 mm × 400 MPa)
= 6.8653 mm
Thread engagement needed against stripping
[54] u_strip = m_min / m_eff
= 6.8653 mm / 12 mm
= 0.57211
Engagement utilisation
[55] F_A_sep = (F_M_min - F_Z) / (1 - Phi) VDI 2230 R7
= (25.431 kN − 3.024 kN) / (1 − 0.072047)
= 24.147 kN
Working load at which the members separate (minimum preload, after embedding)
[56] F_proof = R_p02 * A_s
= 640 MPa × 84.266 mm²
= 53.93 kN
Bolt load at the proof (yield) strength
[57] n_0 = F_A_sep / F_A
= 24.147 kN / 10 kN
= 2.4147
Load factor against separation (F_A,sep / F_A)
[58] F_A_yield = min((F_proof - F_M) / Phi, F_proof)
= min((53.93 kN − 40.69 kN) / 0.072047, 53.93 kN)
= 53.93 kN
Working load at which the bolt reaches its proof load (separation included)
[59] n_L = F_A_yield / F_A
= 53.93 kN / 10 kN
= 5.393
Load factor against axial yield of the bolt, torsion excluded

[60] u_clamp = F_KR_req / F_KR
= 5 kN / 13.128 kN
= 0.38087
Required over available residual clamp

5. Results

SYMBOL	VALUE	DESCRIPTION
n_0	2.4147	Load factor against separation
n_L	5.393	Load factor against axial yield
A_s	84.266 mm²	Tensile stress area
F_M	40.69 kN	Assembly preload (maximum)
F_M_min	25.431 kN	Minimum preload
sigma_M	482.87 MPa	Preload stress
nu_M	0.96866	Assembly utilisation of proof strength
k_S	321190 kN/m	Bolt stiffness
k_P	1907800 kN/m	Members' stiffness
Phi	0.072047	Load factor
F_Z	3.024 kN	Embedding loss
F_SA	0.72047 kN	Additional bolt load
F_KR	13.128 kN	Residual clamp
F_A_sep	24.147 kN	Separation load
F_S_max	41.41 kN	Maximum bolt load
sigma_z_max	491.42 MPa	Maximum bolt tensile stress
sigma_red_B	528.47 MPa	Working equivalent stress
sigma_a	4.2749 MPa	Fatigue amplitude
m_min	6.8653 mm	Engagement needed

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Yield in the working state	u_yield = 0.82574	u_yield ≤ 1	0.826	VDI 2230 R8
PASS	Thread stripping	u_strip = 0.57211	u_strip ≤ 1	0.572	VDI 2230 §5.5.5
PASS	Residual clamp against requirement	u_clamp = 0.38087	u_clamp ≤ 1	0.381	VDI 2230 R7
PASS	Fatigue amplitude	u_fatigue = 0.085499	u_fatigue ≤ 1	0.0855	VDI 2230 R9
PASS	No separation	13.128 kN	> 0 kN	0	VDI 2230 R7

7. Applicability

WARNING — preload-yield: M_A: the preload alone takes the bolt above 90 % of its proof strength.

8. Verification

Template mech.bolted-joint@1.1.0 Content hash efa946ca6a5170d9 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0